

Threat Detection

1. Sign up for a free AWS account (or use an existing one), navigate to AWS GuardDuty in the console, and enable GuardDuty for your region.

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1. Log in to the **AWS Management Console**.
2. Search for **GuardDuty** in the AWS search bar.
3. Open the **Amazon GuardDuty** service.
4. Select your preferred AWS Region (for example, **Asia Pacific – Mumbai**).
5. Click **Get Started** or **Enable GuardDuty**.
6. Confirm the setup.

❖ **Result**

- Amazon GuardDuty is now enabled and continuously monitors your AWS account for suspicious activities using AWS CloudTrail, VPC Flow Logs, DNS logs, and threat intelligence.
2. Simulate a suspicious activity by generating a failed login attempt in your AWS account, then check GuardDuty to see if it detects and logs the event.

***Hint:** Try logging in with the wrong password multiple times or use the AWS CLI to attempt unauthorized actions.*

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1. Open an Incognito/Private browser.
2. Go to the AWS login page.
3. Enter an incorrect password several times for your AWS account.
4. Alternatively, use the AWS CLI with invalid credentials to attempt unauthorized API requests.
5. Wait a few minutes.
6. Open **AWS GuardDuty**.
7. Navigate to **Findings** and review any detected security events.

❖ **Observation**

- If GuardDuty identifies suspicious behavior, it generates a finding with details such as:
- Threat type
 - Severity
 - Affected resource
 - Time of detection
 - Recommended remediation steps

3. In Azure, access the Security Center and enable threat protection for a virtual machine or resource group, then review the dashboard for any security alerts or recommendations.

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1. Log in to the **Azure Portal**.
2. Search for **Microsoft Defender for Cloud** (formerly Azure Security Center).
3. Open **Environment Settings**.
4. Select your subscription.
5. Enable **Defender Plans** for Virtual Machines or Resource Groups.
6. Save the configuration.
7. Open the **Recommendations** or **Security Alerts** dashboard.

❖ **Result**

- Microsoft Defender for Cloud begins monitoring Azure resources and provides:
 - Security recommendations
 - Threat alerts
 - Secure score
 - Compliance reports
- 4. Compare the types of threats detected by AWS GuardDuty and Azure Security Center by listing 3 examples each, based on your exploration of their dashboards and documentation.

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AWS GuardDuty

Detects unusual API calls through AWS CloudTrail.

Detects cryptocurrency mining activity on EC2 instances.

Detects brute-force attacks against EC2 instances, Kubernetes clusters, and other monitored resources.

Azure Security Center (Microsoft Defender for Cloud)

Detects suspicious sign-in attempts and identity-related threats.

Detects malware or suspicious processes running on Azure virtual machines.

Detects network attacks, insecure configurations, and missing security updates.

5. Describe, step-by-step, how real-time threat detection works in either AWS GuardDuty or Azure Security Center, using a scenario like detecting a brute-force attack on a cloud resource.

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1. An attacker repeatedly attempts to log in to an EC2 instance using SSH with different passwords.
2. AWS records the activity through supported telemetry such as VPC Flow Logs and CloudTrail (where applicable).
3. Amazon GuardDuty continuously analyzes these logs using machine learning and AWS threat intelligence.
4. GuardDuty recognizes the repeated login attempts as potential brute-force behavior.
5. A security finding is created with a severity level (Low, Medium, or High).
6. The AWS administrator receives the finding in the GuardDuty console and can integrate notifications with Amazon EventBridge, SNS, or other services.
7. The administrator investigates the affected EC2 instance, blocks the attacker's IP address using security groups or network ACLs, rotates credentials if necessary, and continues monitoring for further suspicious activity.

❖ **Outcome**

- Real-time threat detection helps identify attacks quickly, allowing administrators to respond before attackers gain unauthorized access or cause damage.